

TECHNICAL NOTE

Title:	Lacq Controlled Release Technical Limitations
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1. SUMMARY

Flylogix has been invited to participate in a controlled release experiment at Lacq between 16th - 20th September 2024. The facility where the trial is being conducted has imposed a number of constraints that will impact the quality of the measurement. In the spirit of robust experimentation this document highlights these constraints prior to commencing the campaign.



Fig 1. Flylogix Aircraft conducting methane emission measurement of offshore asset

2. DISCUSSION

Flylogix specialises in conducting asset level emission measurements of offshore oil and gas assets using a long range, fixed wing, drone. By conducting the entire operation using small uncrewed aircraft operating from runways on the coast this reduces the cost, risk and environmental impact of measuring offshore methane emissions. Over the last 6 years Flylogix

has refined an approach where the aircraft flies a cylindrical pattern around the offshore asset forming a closed cylinder whilst making accurate measurements of the local wind, facilitating a mass flow calculation. Flylogix has optimised stand offs to maximise point concentration whilst minimising the impact of wake and plume instabilities. Flylogix use the mass flux method to successfully measure the total emissions down to approximately 1kg/hr with an uncertainty of 2kg/hr. This approach is robust to upwind contamination, plume instability, multiple sources and weather. The method is repeatable, transparent and auditable.

Following successful measurements using this technique of Neptune and TotalEnergies facilities for EDF, Flylogix was encouraged to participate in the TADI trial. Flylogix welcome peer reviewed controlled releases to improve their approach. During planning the Lacq facility have told us that we are not able to conduct our standard technique using the fixed wing uncrewed aircraft and flight patterns deployed offshore due to site restrictions. Flylogix therefore engaged a multirotor drone service provider, Air Control Entech, to fly a reduced diameter cylinder at the Lacq facility. Flylogix believe that the point source and absence of a large superstructure creating a wake mean that we can conduct a meaningful replication of the technique with a smaller diameter cylinder. As the trial approached we have been informed by the Lacq facility that Air Control Entech cannot fly a complete closed circuit or a cylindrical flight path. We have been restricted to flying down wind planes. This invalidates the mass-balance contour integral calculation and introduces a number of risks including upwind contamination and error-prone wind speed and direction measurements as we cannot derive it from instrumentation on our aircraft. Furthermore, mesoscale turbulent fluctuations with timescales comparable to the flight time of the drone can negatively impact flux plane measurements and associated mass-balance calculations at these distances, an issue that is mitigated by the large standoff employed in our standard approach

We remain enthusiastic about participating in the trial as an opportunity to test the sensor but note the calculation methodology and critical wind behaviour is unrepresentative of our proven offshore approach. We will endeavour to mitigate these by making an independent wind measurement, repeating measurements and using alternate calculations but note these will be unrepresentative of our tested methodology. We do have a wealth of experience of performing offshore methane surveys and as such would be very keen to help in defining a more representative test of offshore-monitoring solutions in the future.

3. CONCLUSIONS

Flylogix is keen to participate in the TADI trial.

The constraints on aircraft and flight route at Lacq mean the trial will not be representative of Flylogix's proven offshore measurement methodology.

Flylogix will endeavour to provide a meaningful test of the AERIS Strato instrument by employing an alternative calculation test, using independent wind data and flying repeats.

Long term, Flylogix would welcome the opportunity of defining and participating in a more representative offshore controlled release.